



SYMBIOSIS COLLEGE OF ARTS AND COMMERCE

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FINTECH: BUSINESS STATISTICS WITH R

ACADEMIC YEAR 2026-27

ASSIGNMENT 1: STOCK ANALYSIS USING R

(Deliverables and Interpretations)

STUDENT DETAILS:

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INTRODUCTION

This assignment, part of Business Statistics with R, uses real stock market data to learn practical data analysis skills. Practical work with historical price data from six NSE-listed companies, applying R programming to acquire, clean, and analyze the information.

- **Objective of the Assignment**

The assignment aims to build foundational skills in data manipulation, exploratory data analysis, and statistical modeling using R. Students learn to handle real-world financial data—specifically OHLC (Open, High, Low, Close) prices and trading Volume—for six selected stocks over a five-year period.

- **Selected NSE-listed stocks for analysis:**

1. Telecom Sector: Vodafone Idea Limited – IDEA.NS
2. Telecom Sector: Bharti Airtel Limited – BHARTIARTL.NS
3. Energy Sector: Bharat Petroleum Corporation Limited – BPCL.NS
4. Energy Sector: Oil and Natural Gas Corporation Limited – ONGC.NS
5. Fast-Moving Consumer Goods (FMCG) Sector: Tata Consumer Products Limited – TATACONSUM.NS
6. Fast-Moving Consumer Goods (FMCG) Sector: Hindustan Unilever Limited – HINDUNILVR.NS

- **Rationale for Choosing these Stocks:**

Six NSE stocks span telecom, energy, and FMCG sectors, enabling diverse risk, growth, and market comparisons.

Choosing two stocks per sector allows for **within-sector comparison**, not just across-sector comparison. This helps show that even companies in the same industry can behave differently due to company-specific factors like management, market share, or financial health.

For example:

- **Telecom:** Vodafone Idea (financially stressed, high volatility) vs. Bharti Airtel (market leader, more stable) — shows how firm-specific risk varies within one sector.
- **Energy:** BPCL (downstream/refining) vs. ONGC (upstream/exploration) — shows different exposure to crude oil price movements.

- **FMCG:** Tata Consumer Products vs. Hindustan Unilever — compares a growing/diversifying company with an established market leader.

This pairing strengthens the analysis by testing whether patterns (returns, volatility) are driven by sector-wide trends or by individual company characteristics.

- **Data Acquisition**

1. **Set the time period:** `Sys.Data ( )` captures today's date as the end date, and subtracting 365×5 days gives a start date roughly five years earlier.
2. **Select tickers:** Six NSE stocks are chosen—Vodafone Idea, Bharti Airtel, BPCL, ONGC, Tata Consumer Products, and HUL—each identified by its Yahoo Finance ticker (with ".NS" for NSE).
3. **Download data:** The `getSymbols( )` function from the `qauntmod` package fetches daily OHLCV data for each stock from Yahoo Finance, storing results in separate objects.
4. **Verify data:** Functions like `head ( )`, `str ( )`, and `nrow ( )` check that the data downloaded correctly and show its structure and size.

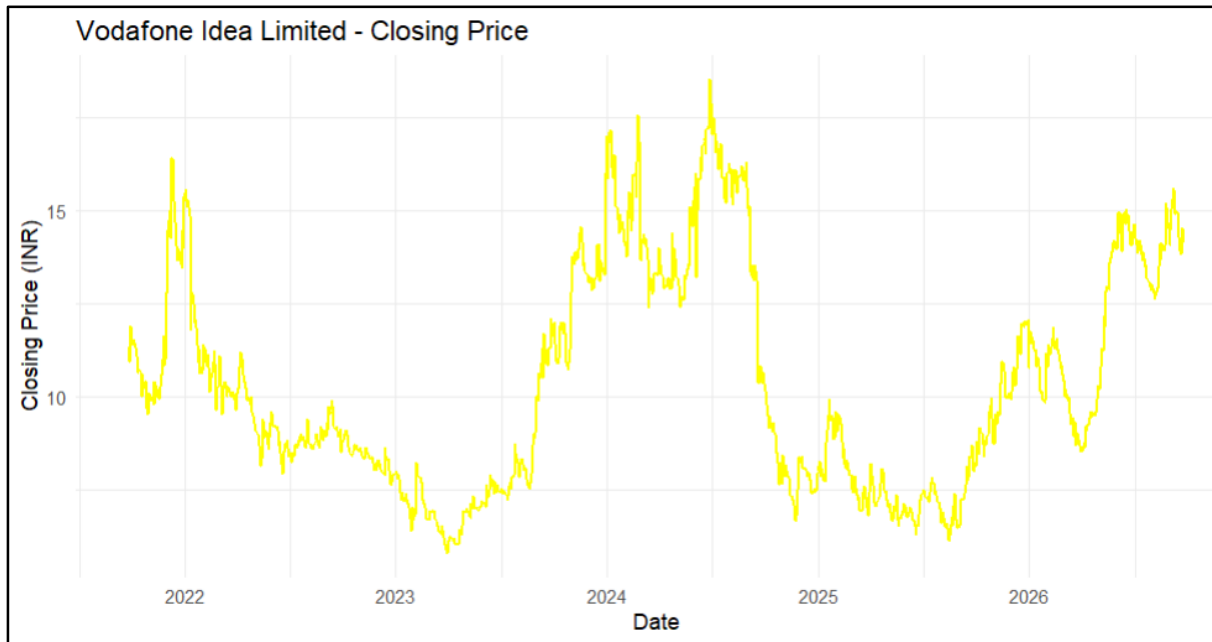
- **Data Cleaning**

1. **Extract closing prices:** `Cl ( )` pulls only the closing price from each stock's data.
2. **Calculate daily returns:** `dailyReturns ( )` computes the percentage change between consecutive days.
3. **Remove missing values:** `na.omit ( )` deletes the first row (which lacks a previous price for comparison).

SECTION A: DESCRIPTIVE STATISTICS

1. LINE PLOTS OF CLOSING PRICES

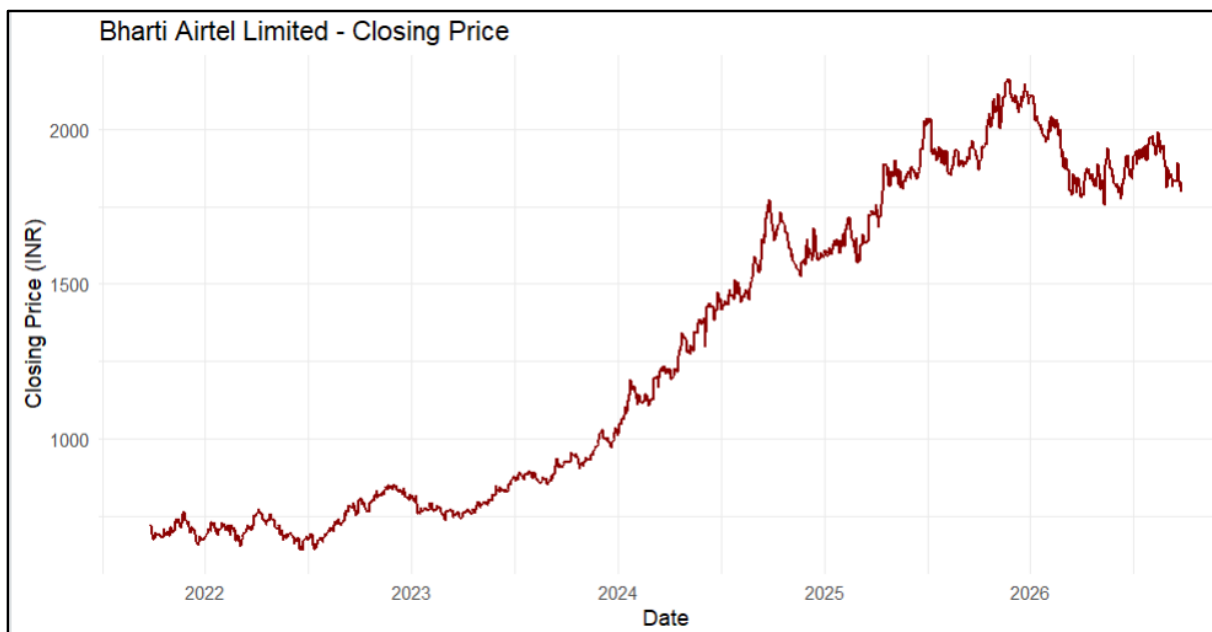
1.1 Vodafone Idea Ltd



Interpretations:

The closing price shows noticeable fluctuations over the period, with some periods of sharp increases and decreases. Overall, the stock appears to be more volatile than some of the other stocks in the dataset.

1.2 Bharti Airtel Ltd



Interpretations:

The closing price shows an overall upward movement over the period, although there are some short-term fluctuations. The graph suggests a relatively strong growth trend in the stock price.

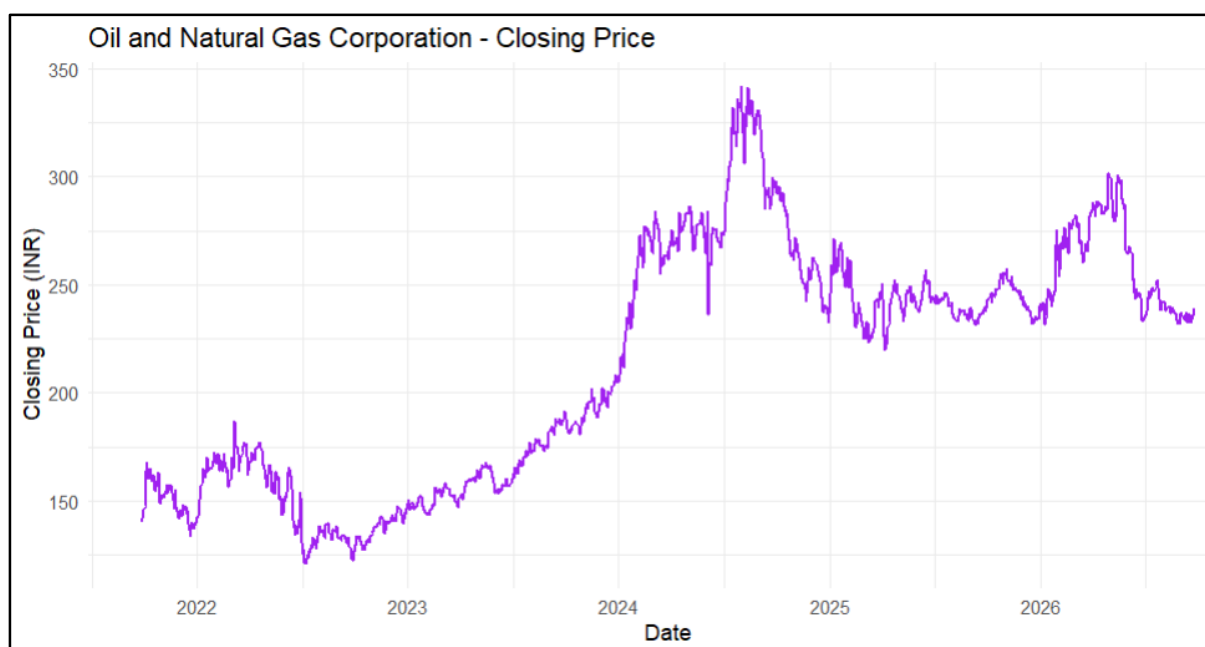
1.3 Bharat Petroleum Corporation Limited



Interpretations:

The closing price shows considerable fluctuations throughout the period, with both upward and downward movements. There are also periods where the price remains relatively stable before changing again.

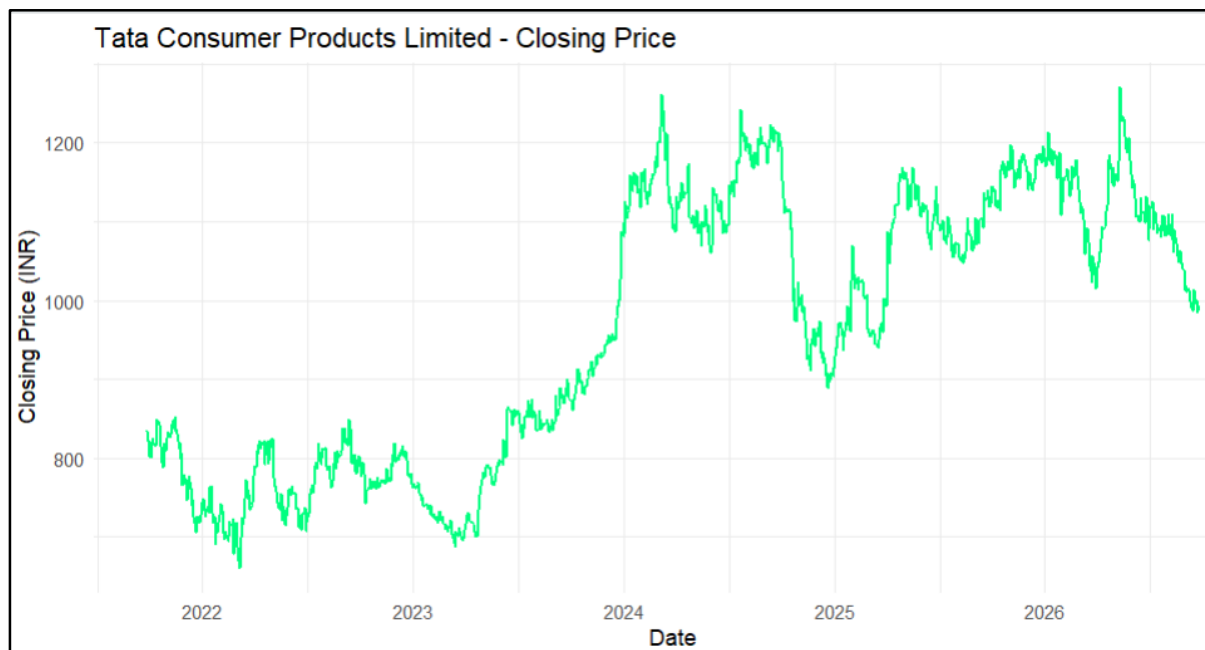
1.4 Oil and Natural Gas Corporation Ltd



Interpretations:

The stock price shows an overall upward movement with several fluctuations during the period. The graph also indicates some periods of relatively sharp price changes after 2024.

1.5 Tata Consumer Products Ltd



Interpretations:

The closing price generally shows an upward trend over the period, although the movement is not continuous. There are several short-term rises and declines in the stock price.

1.6 Hindustan Unilever Ltd



Interpretations:

Hindustan Unilever shows a generally declining trend over the period, despite some temporary increases. The closing price falls noticeably towards the end of the period.

R Output for Closing Prices of all six stocks:

```
> close_statistics
```

	Mean	Median	Standard_Deviation	Variance	Minimum
Vodafone_Idea	10.36827	9.550	2.94969	8.700674	5.800
Bharti_Airtel	1293.38303	1220.400	508.67770	258752.997666	640.900
BPCL	253.33763	265.810	72.60734	5271.826079	146.075
ONGC	213.45271	233.600	55.24334	3051.826439	120.950
TATACONSUM	963.00621	970.575	169.64433	28779.198470	659.850
Hindunilvr	2426.97617	2439.675	198.25158	39303.688994	1927.000

	Maximum
Vodafone_Idea	18.52
Bharti_Airtel	2162.70
BPCL	388.30
ONGC	341.75
TATACONSUM	1271.00
Hindunilvr	3028.55

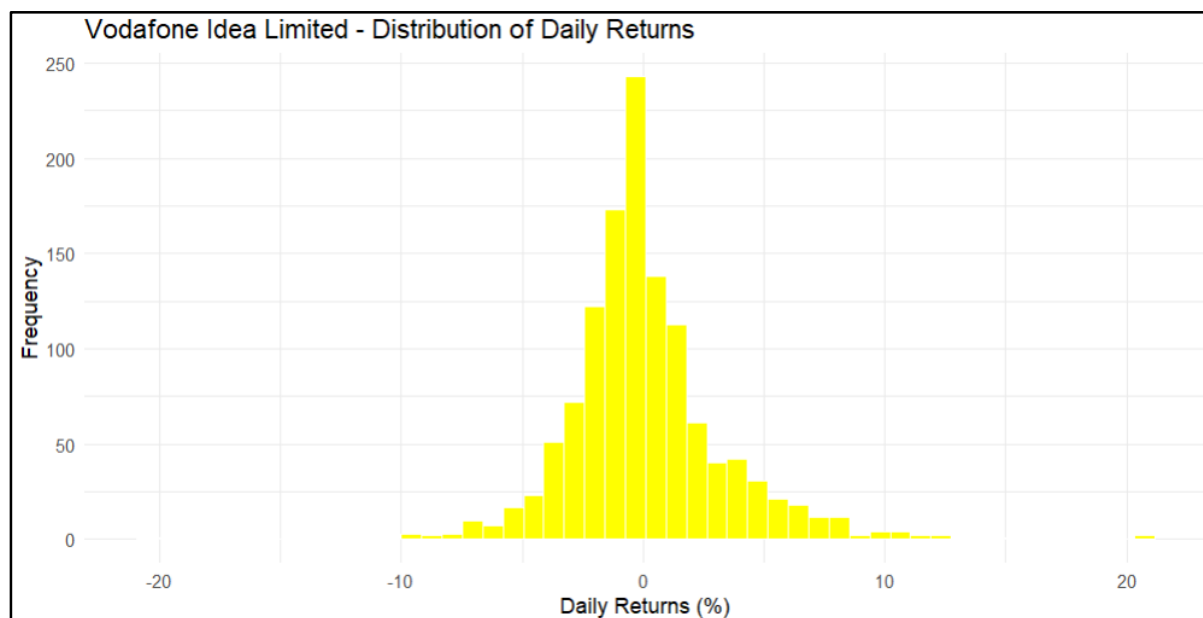
Interpretations:

The mean and median show the typical closing price and are fairly close for most stocks, suggesting no extreme distortion. Standard deviation measures price variability, while variance shows the same spread in squared terms; Bharti Airtel has the largest absolute variation. Minimum and maximum indicate the observed price range, with Vodafone Idea showing the lowest values and Bharti Airtel the highest. Differences in price levels should be considered when comparing variability.

2. HISTOGRAMS OF DAILY RETURNS

The histograms show the distribution of daily percentage returns for the six stocks.

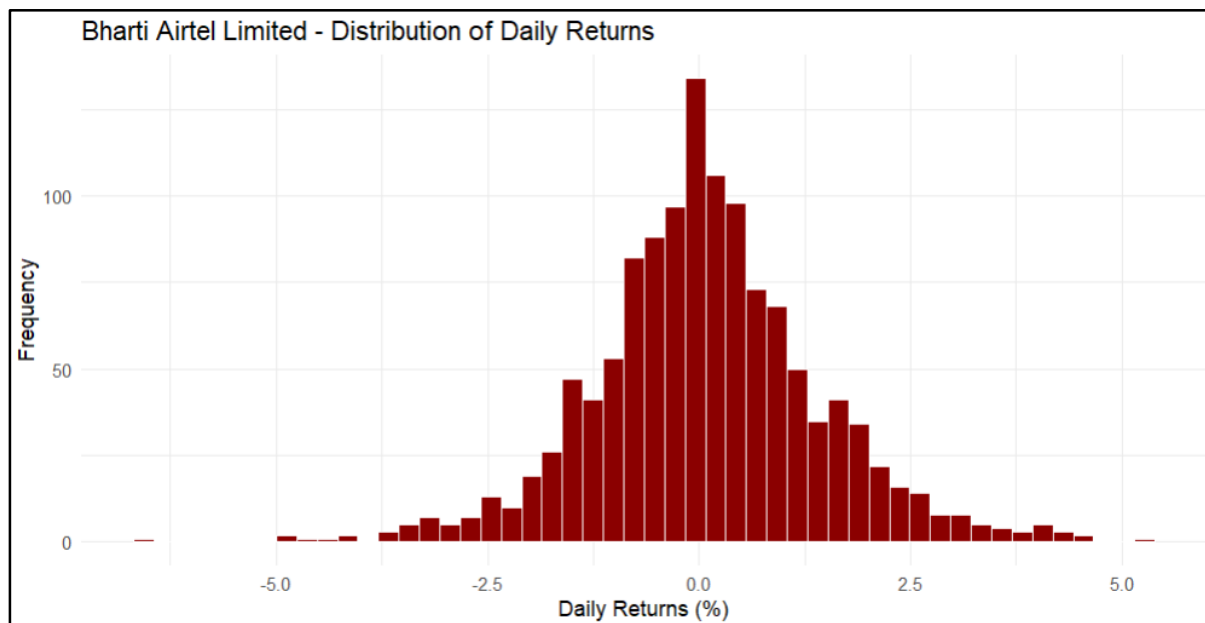
2.1 Vodafone Idea Ltd



Interpretations:

The daily returns are concentrated around zero, with fewer observations at the extreme positive and negative values. The distribution shows that most daily price changes are relatively small.

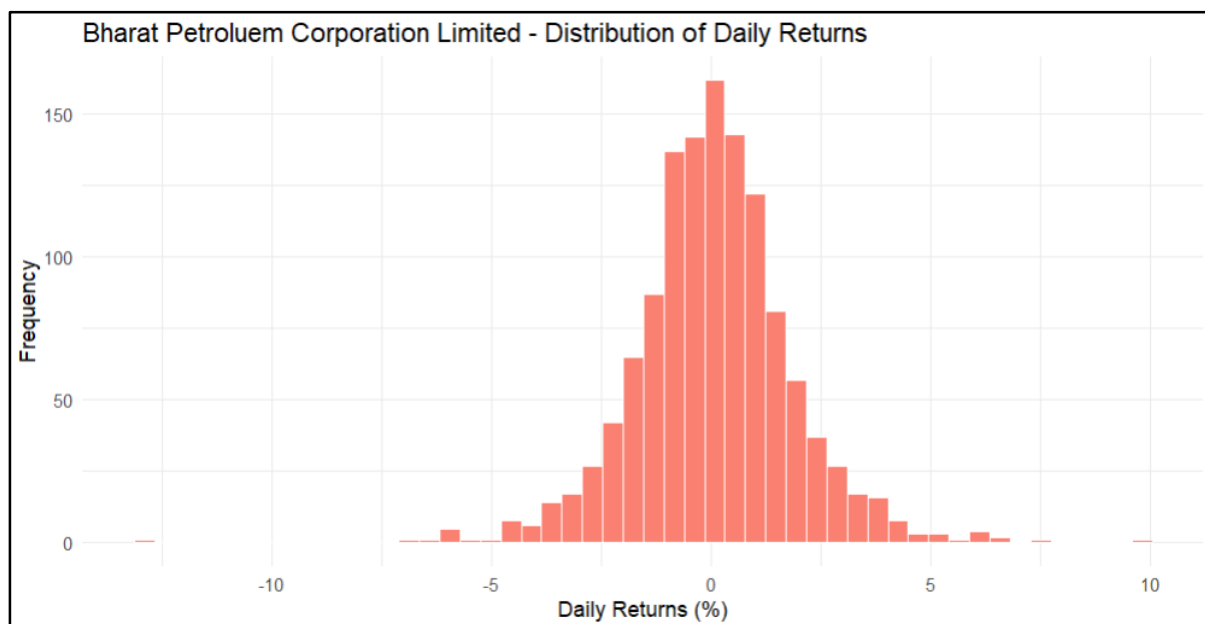
2.2 Bharti Airtel Ltd



Interpretations:

The returns are mainly concentrated around zero and form a fairly bell-shaped distribution. Most daily movements are relatively small, with fewer observations showing larger gains or losses.

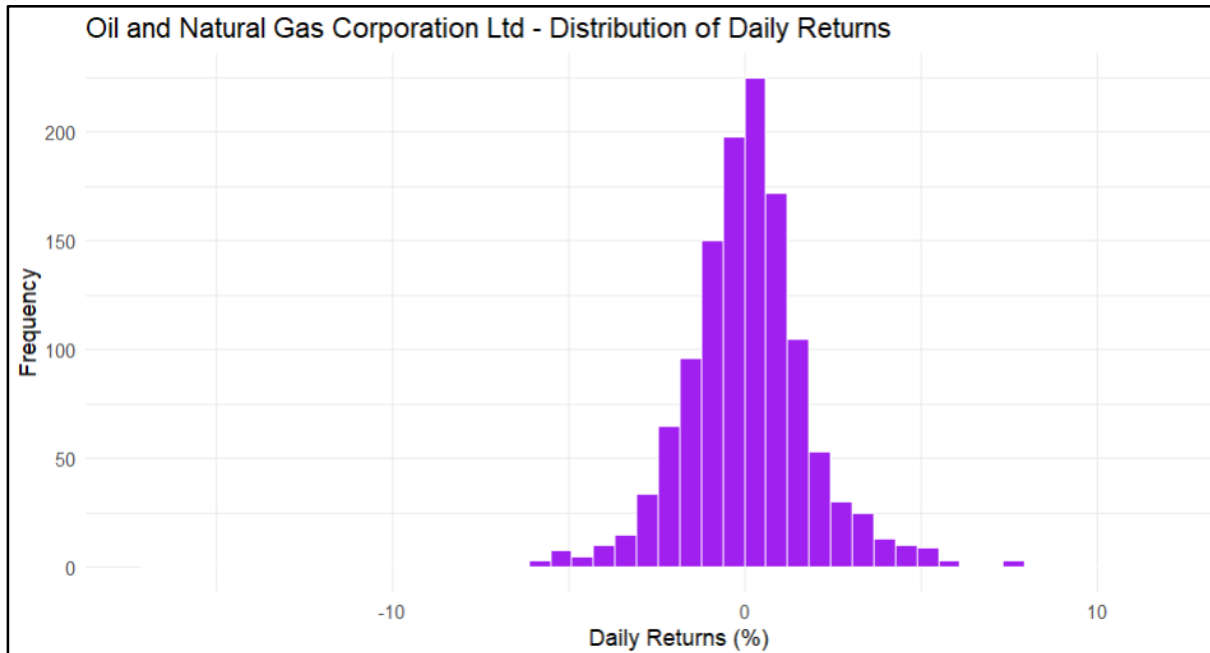
2.3 Bharat Petroleum Corporation Limited



Interpretations:

The histogram shows that most daily returns are centred around zero. There are also some observations in the tails, indicating occasional larger daily price movements.

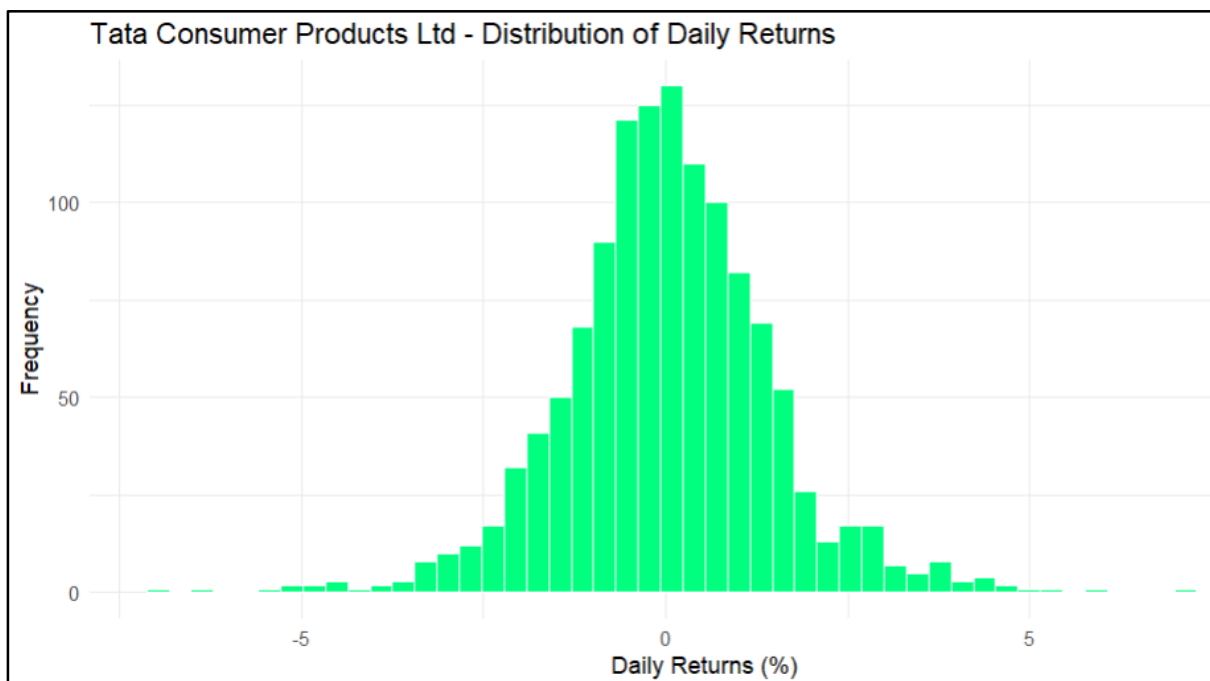
2.4 Oil and Natural Gas Corporation



Interpretations:

The daily returns are mainly concentrated around zero, with fewer observations at higher positive or negative returns. This suggests that most daily changes are relatively small.

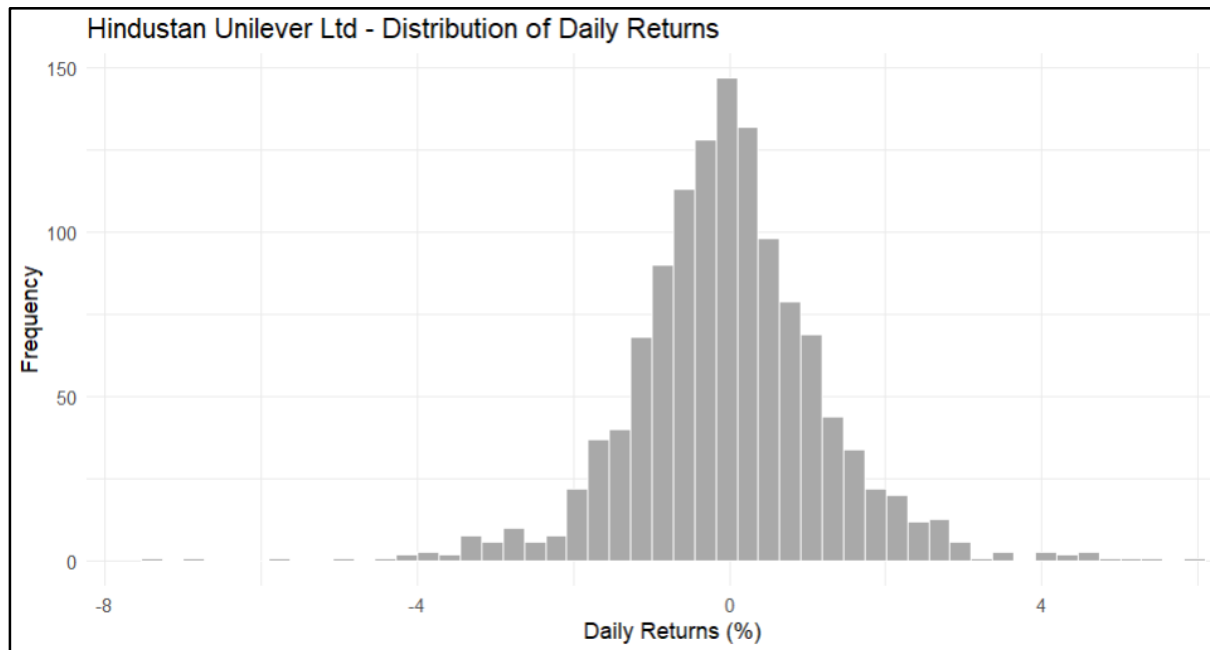
2.5 Tata Consumer Products Ltd



Interpretations:

The daily returns are largely concentrated around zero and show a fairly regular distribution. A smaller number of observations are located in the tails, representing larger daily movements.

2.6 Hindustan Unilever Limited



Interpretations:

The histogram shows a high concentration of daily returns around zero, with relatively few extreme observations.

Overall, the daily return distribution appears to be centred close to zero.

R Output for Daily Returns of all six stocks:

```
> return_statistics
      Mean      Median Standard_Deviation  Variance  Minimum
Vodafone_Idea  0.07508768 -0.26190954      3.405764 11.599227 -20.538721
Bharti_Airtel  0.08260909  0.02496803      1.354562  1.834838  -6.570342
BPCL           0.04696967  0.00000000      1.811784  3.282561 -12.902501
ONGC           0.06198362  0.05004461      1.937326  3.753231 -16.825062
TATACONSUM     0.02423303  0.00000000      1.471640  2.165724  -6.970045
Hindunilvr     -0.01888407 -0.05762437      1.290073  1.664288  -7.400400
      Maximum
Vodafone_Idea 20.754717
Bharti_Airtel  5.270515
BPCL           9.814134
ONGC          13.131617
TATACONSUM     8.059858
Hindunilvr     5.959512
```

Interpretations:

The daily return statistics show that Vodafone Idea has the highest volatility (SD 3.41%, variance 11.60) and the widest range (−20.54% to 20.75%), indicating substantial risk. ONGC also shows relatively high variability, while Bharti Airtel and Hindustan Unilever are comparatively stable. Mean returns are positive for five stocks, with Bharti Airtel highest at 0.083%; Hindustan Unilever alone has a negative mean (−0.019%). Median returns suggest generally modest typical daily movements across the stocks.

SECTION B: REGRESSION ANALYSIS SUMMARY

R output for Regression Summary:

Where,

Dependent Variable: ONGC Daily Returns

Independent Variable: BPCL Daily Returns

Regression Equation:

$$\text{ONGC Return} = 0.0456 + (0.3492 * \text{BPCL Return})$$

```
> summary(regression_model)

Call:
lm(formula = ONGC_Return ~ BPCL_Return, data = regression_data)

Residuals:
    Min       1Q   Median       3Q      Max
-14.6990  -0.9326  -0.0445   0.8794  14.1234

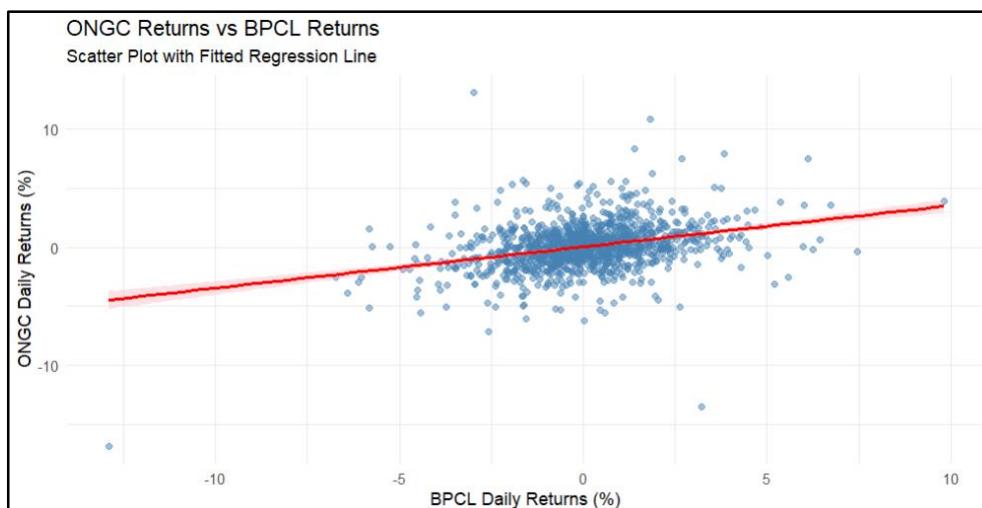
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.04558    0.05204   0.876   0.381
BPCL_Return  0.34916    0.02872  12.155 <0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.832 on 1238 degrees of freedom
Multiple R-squared:  0.1066,    Adjusted R-squared:  0.1059
F-statistic: 147.8 on 1 and 1238 DF,  p-value: < 0.00000000000000022
```

Interpretations:

The regression results show a **positive and statistically significant relationship** between BPCL and ONGC daily returns. The equation indicates that when **BPCL return increases by 1 percentage point, ONGC return is predicted to increase by 0.349 percentage points**, on average. The intercept of 0.0456% is not statistically significant ($p = 0.381$). The slope is highly significant ($p < 0.001$). However, **$R^2 = 0.1066$** means BPCL returns explain only **10.66%** of the variation in ONGC returns. The remaining **89.34%** is explained by other factors.

SCATTER PLOT WITH REGRESSION LINE



Dependent Variable

(Y): ONGC Daily Returns

Independent Variable

(X): BPCL Daily Returns

SECTION C: CROSS-TABULATIONS (CATEGORICAL ANALYSIS)

R output for Cross-Tabulations:

We use Vodafone Idea Ltd and Bharti Airtel Ltd for this section.

Rows = Vodafone Idea Ltd

Columns = Bharti Airtel Ltd

Positive = return > 0

Negative = return < 0

Flat = return = 0

```
> frequency_table
      Bharti_Airtel
Vodafone_Idea Positive Negative Flat
Positive      320      190      3
Negative      276      365      0
Flat          41       38      7
```

```
> round(row_percentages, 2)
      Bharti_Airtel
Vodafone_Idea Positive Negative Flat
Positive      62.38      37.04  0.58
Negative      43.06      56.94  0.00
Flat          47.67      44.19  8.14
```

```
> round(column_percentages, 2)
      Bharti_Airtel
Vodafone_Idea Positive Negative Flat
Positive      50.24      32.04 30.00
Negative      43.33      61.55  0.00
Flat          6.44       6.41 70.00
```

```
> round(total_percentages, 2)
      Bharti_Airtel
Vodafone_Idea Positive Negative Flat
Positive      25.81      15.32  0.24
Negative      22.26      29.44  0.00
Flat          3.31       3.06  0.56
```

Interpretations:

The cross-tabulation shows a **positive association** between Vodafone Idea and Bharti Airtel returns. Both stocks recorded positive returns on **320 days (25.81% of all observations)**. When Vodafone Idea was positive, Bharti Airtel was also positive on **62.38%** of such days. Similarly, both were negative on **365 days (29.44%)**, with Bharti Airtel negative on **56.94%** of Vodafone Idea's negative-return days. Overall, their returns frequently moved in the same direction, indicating some degree of joint movement.

Overall Summary and Key Learnings

- **Descriptive Statistics:** Bharti Airtel, ONGC, and Tata Consumer showed rising price trends, while Hindustan Unilever declined. Vodafone Idea and BPCL were the most volatile. Vodafone Idea had the highest return volatility; Hindustan Unilever had the lowest. Five of six stocks recorded positive average daily returns.
 - **Regression Analysis:** BPCL and ONGC returns showed a positive relationship, consistent with shared exposure to crude oil prices. However, BPCL explains only 10.66% of ONGC's return variation, showing other factors also drive ONGC's performance.
 - **Cross-Tabulation:** Vodafone Idea and Bharti Airtel moved together frequently—both were positive on 25.81% of days and negative on 29.44% of days, showing meaningful joint movement, likely due to shared telecom-sector influences.
 - **Geopolitical Influence:** Events like the West Asia conflict and attacks on Russian energy infrastructure raised crude-price uncertainty (Brent up ~13.4% in early 2024), affecting oil-linked stocks like BPCL and ONGC.
 - **Investment Insight:** Past returns alone don't guarantee reliability—volatility, fundamentals, diversification, and global events matter more.
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